

From a Divergent Newton Series to a Convergent Interpolation of the Primes

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Abstract

We start from the Newton forward-difference series built on the leading entries of the difference table of the primes, $\sum_{n \geq 0} \Delta^n p_1 \binom{x-1}{n}$, and show that it diverges at every non-integer x because $|\Delta^n p_1|$ grows like 2^n . We identify that growth rate as the signature of content at the Nyquist frequency, which places the prime sequence outside the convergence region of the Newton basis. We then track a sequence of repairs. Summation methods fail for a structural reason (a branch point on the Abel path). Recentring the window produces the first convergent form, with residual $O(h^{-1/2})$. Recasting the sliding windows in barycentric form leads to the Floater–Hormann family, and the blending degree $d = 1$ turns out to be exactly the endpoint correction that converts the divergent alternating series in the numerator into a convergent one, via the first step of Euler–Boole summation. The resulting object has a limit as the prime list is extended, the approach to that limit is a single Nyquist mode $c_M \sin(\pi x)$. An exact Abel identity then splits c_M into an analytic part, proved here to be $O(\log M/M)$ unconditionally, and an arithmetic part governed by the alternating sum of prime gaps $G(k) = \sum_{j \leq k} (-1)^j g_j$, bounded by $O(\Phi(M)/M)$ for any regular majorant Φ of $|G|$. The arithmetic part dominates numerically by two to three orders of magnitude, so $c_M = O(1/M)$ is false: it would force $G(k) = O(\log k)$, whereas $|G(2^{18})| = 1.23 \times 10^4$. This also explains quantitatively why Richardson extrapolation in $1/M$ gains about three digits and then stalls. We then reduce the rate question to square-root cancellation in $\sum_{j \leq k} (-1)^j (g_j - \log p_j)$, equivalently in the alternating sum of the primes; show that the Riemann Hypothesis controls the unsigned counterpart of this sum and nothing else; measure G to be diffusive with no drift, its per-step mean square matching $\text{Var}(g)$ to five figures; and record the resulting conditional target $c_M \ll M^{-1/2} \log M \sqrt{\log \log M}$. All claims are accompanied by the numerical checks that support them, and Section 12 states plainly what is established and what is not.

1 The object

Let $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ and let Δ be the forward difference on the index, $\Delta p_k = p_{k+1} - p_k$. Writing $\Delta^n p_c$ for the c -th entry of the n -th row of the difference table, the Newton forward-difference series based at $c = 1$ is

$$N_h(x) = \sum_{n=0}^h \Delta^n p_1 \binom{x-1}{n}, \quad \Delta^n p_1 = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} p_{j+1}. \quad (1)$$

At integers $x = 1, \dots, h+1$ this reproduces p_x exactly, so it is a candidate interpolation of the primes to real argument. The first twenty leading differences are

$$\begin{aligned} &2, 1, 1, -1, 3, -9, 23, -53, 115, -237, \\ &457, -801, 1213, -1389, 445, 3667, -15081, 41335, -95059, 195769. \end{aligned}$$

The question of this note is whether $h \rightarrow \infty$ in (1) produces anything, and if not, what does.

2 Why the series diverges

2.1 Growth of the leading differences

Computing the full difference table of the first 250 primes, the leading entries alternate in sign from $n = 3$ onward with only seven exceptional adjacent pairs, at (14, 15), (25, 26), (38, 39), (46, 47), (97, 98), (111, 112) and (175, 176), and their magnitudes grow geometrically. The consecutive ratios over the last five rows are

$$2.0521, \quad 2.0483, \quad 2.0451, \quad 2.0426, \quad 2.0407, \quad 2.0393,$$

drifting slowly downward. The final entry $\Delta^{249} p_1$ has 75 digits. Removing the anomalous first gap by restarting the table at $p_2 = 3$ changes nothing: the corresponding ratios are $2.0447, \dots, 2.0380$. So we take

$$|\Delta^n p_1| \sim C \rho^n, \quad \rho \approx 2. \quad (2)$$

That $\rho \approx 2$ is not special to the primes. For a sequence with random-like content at the highest representable frequency, $\Delta^n f_1 = \sum_j (-1)^{n-j} \binom{n}{j} f_{j+1}$ has root-mean-square size

$$\left(\sum_j \binom{n}{j}^2 \right)^{1/2} = \binom{2n}{n}^{1/2} \approx 2^n (\pi n)^{-1/4},$$

so 2^n is the generic ceiling. The primes sit essentially at it.

2.2 Asymptotics of the terms

Using the reflection identity

$$\binom{x-1}{n} = (-1)^n \binom{n-x}{n} = (-1)^n \frac{\Gamma(n-x+1)}{\Gamma(n+1)\Gamma(1-x)} \sim \frac{(-1)^n n^{-x}}{\Gamma(1-x)}, \quad (3)$$

and combining with (2), the n -th term of (1) behaves like

$$\Delta^n p_1 \binom{x-1}{n} \sim \frac{C \rho^n n^{-x}}{\Gamma(1-x)}. \quad (4)$$

The two sign alternations cancel. The terms are eventually of one sign and grow geometrically, so there is no cancellation available and the series diverges at every non-integer x .

This is confirmed directly. Using 250 leading differences, every term from $n = 200$ to $n = 205$ is positive at $x = 1.5$ and negative at $x = 2.5$ and $x = 10.5$, with consecutive ratios at $n = 249$ of 2.027, 2.019 and 1.953 respectively. Figure 1(a) shows the term magnitudes climbing on a straight line of slope $\log_{10} 2$.

Two structural facts fall out of (4). The factor $1/\Gamma(1-x)$ has poles at $x = 1, 2, 3, \dots$, so all terms past $n = x - 1$ vanish identically at the integers, which is why the series is trivially exact there. And the sign of $1/\Gamma(1-x)$ flips on each successive unit interval, which is why the truncated sums blow up alternately upward and downward as x crosses each integer.

Remark 1. This is not Runge's phenomenon. Runge concerns a fixed analytic function sampled on equispaced nodes as the degree grows, and its standard cure is to cluster the nodes in a Chebyshev distribution. Here the nodes are the positive integers and cannot be moved, because they are where the data lives. The mechanism is the growth of the divided differences, not the geometry of a fixed interval.

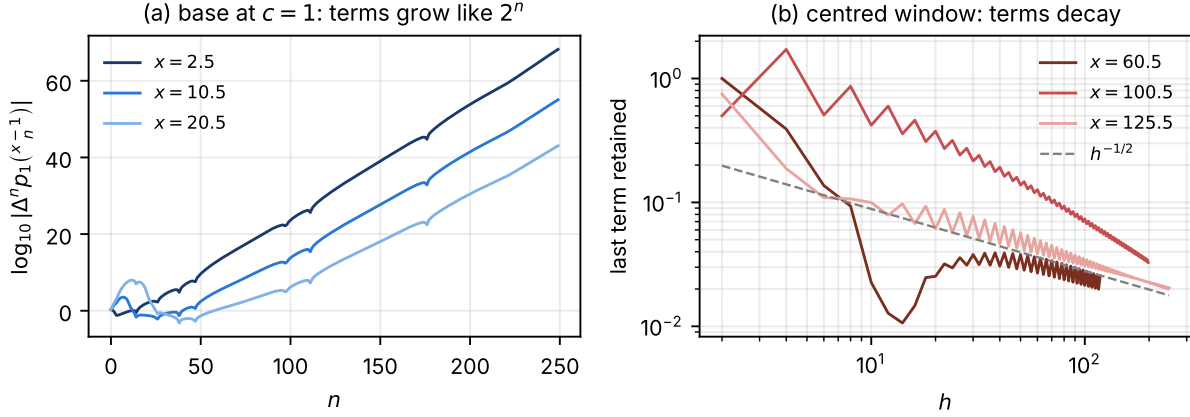


Figure 1: (a) Term magnitudes of (1) at three abscissas, with the base at $c = 1$: after an initial dip the terms grow geometrically at rate ≈ 2 . (b) The last term retained by the centred scheme of Section 4, against window width h : the same data, reorganised, now decays like $h^{-1/2}$.

2.3 The bandwidth reading

Growth rate ρ has a spectral meaning. Applying Δ^n to a mode $e^{i\omega k}$ gives $(e^{i\omega} - 1)^n$, of magnitude $|2 \sin(\omega/2)|^n$. For a sequence band-limited to $|\omega| \leq \tau$ the difference growth is therefore at most $2 \sin(\tau/2)$, an increasing function of τ on $[0, \pi]$ with maximum value 2 attained exactly at the Nyquist frequency $\tau = \pi$. The measured $\rho \rightarrow 2$ says the prime sequence saturates the band.

The Newton basis, by contrast, converges only for data extending to a function of exponential type below $\ln 2 \approx 0.693$ (Nörlund). The gap is a factor of $\pi / \ln 2 \approx 4.5$. The divergence is therefore not an artefact of truncation order or of the base point; it is a mismatch between the bandwidth of the data and the bandwidth the basis can represent. Any repair must either change the basis or change what is meant by summation.

3 Two repairs that do not work

3.1 Summation methods

Since the terms are eventually geometric with ratio near 2, Shanks or Wynn acceleration ought to sum them the way it assigns -1 to $\sum 2^n$. It does not. On the clean model problem $f_k = (-1)^{k-1}$, whose leading differences are exactly $(-2)^n$ and whose natural continuation is $\cos(\pi(x-1))$, the ε -algorithm returns -5.6×10^{81} at $x = 1.5$ and -0.404 at $x = 2.25$ against the true -0.707 .

The obstruction is structural rather than numerical. The generating function is

$$\sum_{n \geq 0} (-2)^n \binom{x-1}{n} z^n = (1-2z)^{x-1}, \quad (5)$$

which has a branch point at $z = \frac{1}{2}$. The Abel path from $z = 0$ to $z = 1$ runs into the cut. The two branches give $e^{\pm i\pi(x-1)}$, so no real Abel sum exists; the best available statement is that their average is $\cos(\pi(x-1))$. A real-valued summation method applied on the real axis is being asked for an object that is not there.

3.2 Optimal truncation

Treating (1) as an asymptotic series and stopping at the least term does work, in the sense of producing a bounded curve. The least term is 2.1×10^{-2} at $x = 4.5$ and 6.1×10^{-4} at $x = 19.5$, and the resulting values (9.815 at $x = 4.5$, 20.411 at $x = 8.5$, 39.596 at $x = 12.5$) sit sensibly between the bracketing primes. But the accuracy is capped by the least term and the curve jumps where the optimal order switches, by 1.62 at $x \approx 3.24$ and 0.70 at $x \approx 9.59$. It is a superasymptotic answer, not a convergent one.

4 Recentring: the first convergent form

Re-base the series at an arbitrary c ,

$$N_{c,h}(x) = \sum_{n=0}^h \Delta^n p_c \binom{x-c}{n}, \quad (6)$$

which interpolates $p_c, \dots, p_{c+h}$. A table built from N primes has $N - n$ entries in row n , so the c -th entry exists precisely when $n \leq N - c$; the truncation bound $h \leq N - c$ is forced by the data rather than chosen.

The crucial quantity is where x sits inside the window. Writing $x - c = h/2 + t$ with $t \in [0, 1)$,

$$\left| \binom{x-c}{h+1} \right| = \frac{1}{(h+1)!} \prod_{j=0}^h \left| \frac{h}{2} + t - j \right| \approx \frac{((h/2)!)^2}{(h+1)!} \sim \sqrt{\frac{\pi h}{2}} \frac{2^{-h}}{h}. \quad (7)$$

The 2^{-h} here cancels the ρ^h of (2) exactly in the limit $\rho \rightarrow 2$, leaving a residual of size $O(h^{-1/2})$. Centring converts divergence into convergence.

The cancellation is dramatic. At $x = 125.5$ with $c = 2$ and $h = 248$,

$$\Delta^{248} p_2 \cdot \binom{123.5}{248} = (1.828 \times 10^{74}) \times (1.119 \times 10^{-76}) = 0.020461,$$

about 150 orders of magnitude of cancellation landing on a small number. Figure 1(b) shows the last retained term following $h^{-1/2}$ once $h \gtrsim 30$. Converged values include 281.8389 at $x = 60.5$, 545.5067 at $x = 100.5$ and 695.8404 at $x = 125.5$.

Two defects remain. First, $O(h^{-1/2})$ is slow: with 250 primes one gets about two decimal places, and each further digit costs a hundredfold increase in h . Second, the construction is not canonical. If c is chosen per x , say $c = \text{round}(x - h/2)$, the curve is discontinuous where c increments. That jump has a closed form: two adjacent windows agree at h common nodes, so their difference is $A \prod_{j=1}^h (x - c - j)$ with $A = \Delta^{h+1} p_c / h!$, giving at the switch point $x^* = c + \frac{h+1}{2}$

$$J = \frac{\Delta^{h+1} p_c}{h!} \prod_{j=1}^h \left(\frac{h+1}{2} - j \right). \quad (8)$$

Evaluated at $c = 60$ this reproduces the measured jumps exactly: 0.23724537 at $h = 20$, 0.15517069 at $h = 40$, -0.17773625 at $h = 80$, -0.061394627 at $h = 120$. Note that these are the same order as the truncation residual. The discontinuity is not an artefact of the switching rule; it is a faithful readout of how much the interpolant is undetermined.

5 From windows to weights

The natural way to remove the switching is to trust each window in inverse proportion to its own error factor, that is, to weight window i by the reciprocal of its node polynomial. Doing this for all sliding windows of a fixed length simultaneously is exactly the Floater–Hormann barycentric rational interpolant. With blending degree d and local polynomials P_i through nodes $x_i, \dots, x_{i+d}$,

$$r(x) = \frac{\sum_i \lambda_i(x) P_i(x)}{\sum_i \lambda_i(x)}, \quad \lambda_i(x) = \frac{(-1)^i}{(x - x_i) \cdots (x - x_{i+d})}. \quad (9)$$

For equispaced nodes this collapses to the barycentric form

$$r_M(x) = \frac{\sum_{k=1}^M \frac{w_k}{x - k} p_k}{\sum_{k=1}^M \frac{w_k}{x - k}}, \quad w_k = (-1)^k \sum_{j=\max(0, k+d-M)}^{\min(d, k-1)} \binom{d}{j}, \quad (10)$$

(the overall sign convention is immaterial, since it cancels in the quotient). This object is C^∞ with no poles on the real line, interpolates exactly at every node, and has a Lebesgue constant growing linearly rather than like 2^h , which removes the Runge behaviour outright.

For $d \leq k \leq M - d$ the inner sum is the full binomial total, so

$$w_k = (-1)^k 2^d \quad \text{throughout the interior.} \quad (11)$$

Only the first d and last d weights depend on d at all. Numerically, $r(60.5)$ moves from 281.88161 at $d = 1$ to 281.89735 at $d = 12$, five significant figures of agreement, while the edge values move enormously: $r(2.5)$ runs from 4.238 at $d = 1$ to 25.43 at $d = 12$. The degree is an edge parameter.

Choosing $d = 1$ minimises overshoot past the bracketing primes (0.533, against 0.612 at $d = 0$, 17.50 at $d = 9$ and 145.03 at $d = 12$) and ties for lowest curvature. At $d = 1$ the weights are simply

$$w_k = (-1)^k \times \begin{cases} 1, & k = 1 \text{ or } k = M, \\ 2, & \text{otherwise.} \end{cases} \quad (12)$$

The endpoints are halved. Section 6 shows that this halving is not cosmetic.

6 Why $d = 1$ converts divergence into convergence

Dividing (12) through by 2, the numerator and denominator of (10) are

$$N_M(x) = \sum_{k=1}^M \frac{(-1)^k p_k}{x - k} - \frac{1}{2} \left(\frac{-p_1}{x-1} + \frac{(-1)^M p_M}{x-M} \right), \quad D_M(x) = \sum_{k=1}^M \frac{(-1)^k}{x - k} - \frac{1}{2} \left(\frac{-1}{x-1} + \frac{(-1)^M}{x-M} \right). \quad (13)$$

6.1 The denominator converges, and in closed form

The terms of D_M decay like $1/k$ with alternating signs, so it converges. Splitting the Mittag-Leffler expansion $\pi/\sin(\pi x) = \sum_{k \in \mathbb{Z}} (-1)^k/(x - k)$ at $k = 0$ gives

$$D_\infty(x) = \frac{\pi}{\sin \pi x} - \frac{1}{2} \left[\psi\left(\frac{x+1}{2}\right) - \psi\left(\frac{x}{2}\right) \right] + \frac{1}{2(x-1)}, \quad (14)$$

where the middle term is $\sum_{m \geq 0} (-1)^m / (x + m)$ and the last is the halved first weight. At $x = 60.5$ the three pieces are 3.14159265, -0.00833276 and $+0.00840336$, totalling 3.14166326, against the numerically summed 3.1416632601.

Expanding the last two pieces, $\frac{1}{2(x-1)} = \frac{1}{2x} + \frac{1}{2x^2} + O(x^{-3})$ and $\sum_{m \geq 0} \frac{(-1)^m}{x+m} = \frac{1}{2x} + \frac{1}{4x^2} + O(x^{-3})$, so they very nearly cancel and

$$D_\infty(x) = \frac{\pi}{\sin \pi x} + \frac{1}{4x^2} + O(x^{-3}). \quad (15)$$

At $x = 60.5$ the predicted correction is 6.8×10^{-5} and the observed one is 7.1×10^{-5} . Equation (15) is the key to Section 7.

6.2 The numerator diverges, and the halving repairs it

The numerator is the interesting one. Since $p_k \sim k \ln k$,

$$\frac{(-1)^k p_k}{x - k} \rightarrow -(-1)^k \ln k, \quad (16)$$

so the terms do *not* tend to zero. As a series, $\sum_k (-1)^k p_k / (x - k)$ diverges outright. What rescues it is precisely the endpoint halving in (12).

For an alternating sum with slowly varying a_k , Euler–Boole summation gives

$$\sum_{k=1}^M (-1)^k a_k = C + \frac{(-1)^M a_M}{2} + \frac{(-1)^M a'_M}{4} + \dots, \quad (17)$$

an asymptotic expansion whose leading oscillatory term is exactly half the last term. Subtracting that half term removes the leading oscillation and leaves a residual of size $O(a'_M) = O(1/M)$. Comparing (13) with (17), the $d = 1$ weight pattern is the first step of Euler–Boole summation, applied automatically by the interpolation scheme.

The numbers make this unambiguous. At $x = 60.5$, with $M = 10^3, 10^4, 10^5, 10^6$:

M	10^3	10^4	10^5	10^6
N_M with the halving	885.227	885.269	885.150	885.125
N_M without it	880.996	879.984	878.631	877.365

The first row settles; the second drifts without limit, losing about 1.3 per decade, which is the $\frac{1}{2} \ln M$ growth predicted by (17). The gap between the rows at $M = 10^6$ is 7.7602, against the predicted half-term $\frac{1}{2} p_{10^6} / (10^6 - x) = \frac{1}{2} \cdot 15485863 / 999939.5 = 7.7434$, the small remainder being the halved first term $p_1 / (2(x - 1)) = 0.0168$.

This is the pivot of the whole construction. A divergent alternating series in the numerator is divided by a convergent one in the denominator, and a single endpoint correction, supplied for free by the choice $d = 1$, makes the numerator converge as well.

Remark 2. The choice $d = 0$, Berrut’s original interpolant, does *not* halve the endpoints, so it does not carry the Euler–Boole correction. Correspondingly $r(60.5) = 281.620$ at $d = 0$ against 281.882 for every $d \geq 1$. Degrees $d \geq 2$ also halve the endpoints, in a more elaborate way, which is why they converge to the same interior limit. So $d = 1$ is the minimal choice that works, not the only one.

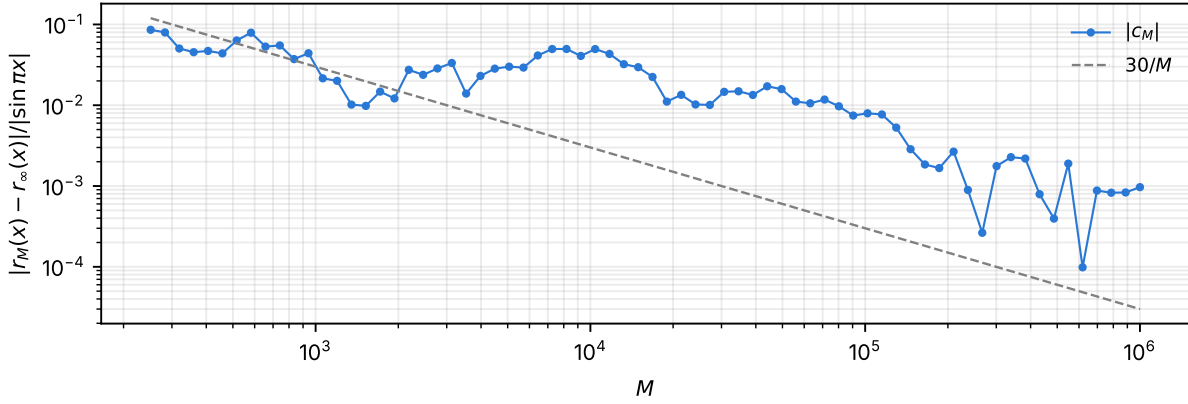


Figure 2: Amplitude of the residual Nyquist mode against the number of primes used, with a $1/M$ reference. The scatter is not measurement noise. Section 8 identifies it as an arithmetic term that dominates the smooth one.

7 The residual is a single Nyquist mode

For $k \gg x$ one has $1/(x - k) = -1/k + O(x/k^2)$, so the tail contributions to N_M and D_M are independent of x to leading order. Writing $N_M = N_\infty + \delta_N$ and $D_M = D_\infty + \delta_D$ with δ_N, δ_D approximately x -free,

$$r_M(x) - r_\infty(x) \approx \frac{\delta_N - r_\infty \delta_D}{D_\infty(x)} \approx c_M \sin(\pi x), \quad (18)$$

the last step by (15). The prediction is sharp: the entire dependence on how many primes are used is one scalar times the Nyquist mode.

Solving (18) for c_M at eight abscissas confirms it.

x	60.15	60.25	60.5	60.75	60.85	61.5	124.3	125.5
$c_M, M = 10^4$	0.04575	0.04575	0.04575	0.04575	0.04575	0.04575	0.04591	0.04592
$c_M, M = 10^5$	0.00787	0.00787	0.00787	0.00787	0.00787	0.00787	0.00787	0.00787

The value is constant to five decimals across different integer cells, different fractional offsets, and both signs of $\sin(\pi x)$. The slight drift in the last two columns is the $1/(4x^2)$ term of (15), of the predicted size.

Figure 2 shows $|c_M|$ decaying, so the limit exists. The decay is irregular and no power law should be read off it by eye; Section 8 determines the rate properly.

Remark 3 (on a discarded estimate). An earlier version of these notes fitted $\log |c_M|$ against $\log M$ pointwise over $M \in [3 \times 10^3, 6 \times 10^5]$, obtained an exponent of -0.942 , and read it as confirming $c_M = O(1/M)$. That estimator is not valid here. Section 8 shows that c_M contains a term behaving like a random walk, so individual values are not representative of their local scale and a least-squares line through them largely records where the walk happened to sit at the ends of the fitting window. Taking instead the root mean square of $|c_M|$ over dyadic blocks gives an exponent of -0.38 , with block-to-block ratios ranging from 0.64 to 3.66. The honest reading of that is not “the exponent is -0.38 ” but “no exponent is determined by data over this range”, which is what one should expect once the dominant term is arithmetic rather than analytic.

8 The rate: what can be proved

Throughout this section x is fixed and non-integral, $a_k = p_k/(x - k)$, $g_k = p_{k+1} - p_k$, and implied constants may depend on x .

8.1 An exact half-sum identity

Lemma 4. *For any sequence (a_k) and any $M \geq 2$,*

$$\sum_{k=1}^M (-1)^k a_k - \frac{1}{2} \left((-1)a_1 + (-1)^M a_M \right) = -\frac{1}{2} \sum_{k=1}^{M-1} (-1)^k (a_{k+1} - a_k). \quad (19)$$

Proof. Put $A_k = (-1)^k/2$, so that $A_k - A_{k-1} = (-1)^k$. Abel summation gives $\sum_{k=1}^M (A_k - A_{k-1})a_k = A_M a_M - A_0 a_1 - \sum_{k=1}^{M-1} A_k (a_{k+1} - a_k)$. Substituting $A_M = (-1)^M/2$ and $A_0 = \frac{1}{2}$ and moving the two boundary terms to the left gives (19). $\square$

The left side of (19) is exactly the numerator $N_M(x)$ of (13). So the $d = 1$ scheme does not merely approximate an Euler–Boole correction; it is identically equal to a half-sum of first differences. Checked against direct evaluation at $x = 60.5$, the two sides agree to ten decimals at $M = 10, 10^2, \dots, 10^6$, for instance 885.1252016789 against 885.1252016788 at $M = 10^6$. Consequently

$$N_\infty(x) - N_M(x) = -\frac{1}{2} \sum_{k \geq M} (-1)^k \Delta a_k, \quad (20)$$

and the whole question of the rate is the size of that tail.

8.2 Splitting the tail

A direct computation gives the exact decomposition

$$\Delta a_k = \underbrace{\frac{g_k}{x - k - 1}}_{\text{arithmetic}} + \underbrace{\frac{p_k}{(x - k - 1)(x - k)}}_{\text{analytic}}, \quad (21)$$

verified numerically to 5.7×10^{-14} over $k \leq 10^6$. The names are chosen because the first term carries the irregularity of the prime gaps while the second depends on p_k only through its smooth size.

Proposition 5 (analytic part). *Let $\phi_k = p_k/((k+1-x)(k-x))$. Then $|\sum_{k \geq M} (-1)^k \phi_k| \ll_x \log M/M$.*

Proof. Since the partial sums of $(-1)^k$ are bounded by 1 and $\phi_k \rightarrow 0$, Abel summation bounds the tail by the total variation $\sum_{k \geq M} |\phi_{k+1} - \phi_k|$. Clearing denominators,

$$\phi_{k+1} - \phi_k = \frac{g_k(k-x) - 2p_k}{(k+2-x)(k+1-x)(k-x)} \ll \frac{g_k}{k^2} + \frac{p_k}{k^3}.$$

For the second term, $p_k \ll k \log k$ gives $\sum_{k \geq M} p_k/k^3 \ll \sum_{k \geq M} \log k/k^2 \ll \log M/M$. For the first, partial summation with $\sum_{k \leq K} g_k = p_{K+1} - p_1 \ll K \log K$ gives

$$\sum_{k \geq M} \frac{g_k}{k^2} \ll \frac{M \log M}{M^2} + \sum_{k \geq M} \frac{k \log k}{k^3} \ll \frac{\log M}{M}. \quad \square$$

Only Chebyshev-type bounds are used, so Proposition 5 is unconditional.

Proposition 6 (arithmetic part). *Let $G(k) = \sum_{j \leq k} (-1)^j g_j$ and suppose Φ is nondecreasing with $|G(k)| \leq \Phi(k)$ and $\Phi(k) = k^\theta L(k)$ for some $\theta < 1$ and slowly varying L . Then*

$$\left| \sum_{k \geq M} \frac{(-1)^k g_k}{k+1-x} \right| \ll_x \frac{\Phi(M)}{M}.$$

Proof. Write $\psi_k = (k+1-x)^{-1}$, which decreases to 0 with $\psi_k - \psi_{k+1} \ll k^{-2}$. Abel summation against the partial sums $G(k) - G(M-1)$ gives

$$\left| \sum_{k \geq M} (-1)^k g_k \psi_k \right| \leq 2\Phi(M)\psi_M + 2 \sum_{k \geq M} \Phi(k) (\psi_k - \psi_{k+1}) \ll \frac{\Phi(M)}{M} + \sum_{k \geq M} \frac{\Phi(k)}{k^2},$$

and $\theta < 1$ makes the last sum $\ll \Phi(M)/M$. □

Proposition 7. *With Φ as above, $|c_M| \ll_x \frac{\log M}{M} + \frac{\Phi(M)}{M}$.*

Proof. Combine (20), (21), Propositions 5 and 6, and divide by $D_\infty(x)$, which is bounded away from 0 for fixed non-integral x by (15). The corresponding tail for D_M is the special case $p_k \equiv 1$ of (21), for which the arithmetic term is absent and Proposition 5 applies directly. □

8.3 Why $c_M = O(1/M)$ is false

Proposition 7 reduces the rate to the growth of the alternating gap sum G . Reading it backwards: $c_M = O(1/M)$ would require $\Phi(M) = O(1)$, or with the $\log M/M$ term absorbed, $G(k) = O(\log k)$. That is essentially total cancellation in $\sum_{j \leq k} (-1)^j g_j$, and it is contradicted by the data. Figure 3(b) plots $|G(k)|$: it reaches 1.23×10^4 at $k = 2^{18}$, against $\log k \approx 12.5$. The root mean square of $|G|$ over dyadic blocks tracks $\sqrt{k} \log k$ within a factor of two, drifting from 0.38 times that reference at $k \approx 2^8$ to 1.56 at $k \approx 2^{18}$, which is the behaviour of a quantity with square-root cancellation sitting in a long one-signed excursion, not of a bounded one.

Corollary 8. *If the alternating gap sum exhibits square-root cancellation, $G(k) \ll k^{1/2+\varepsilon}$, then $c_M \ll_{x,\varepsilon} M^{-1/2+\varepsilon}$.*

That is the strongest clean statement available, and it is conditional. Unconditionally, the trivial bound $|G(k)| \leq p_{k+1}$ gives only $c_M \ll \log M$, which says nothing.

The split is also directly visible. Evaluating the two tails of (21) separately at $x = 60.5$:

M	analytic tail	$\log M/M$	arithmetic tail
10^3	+0.002224	0.006908	-0.102949
10^4	+0.000265	0.000921	-0.143335
10^5	+0.000036	0.000115	-0.023918
3×10^5	+0.000016	0.000042	-0.001191

The analytic tail sits at about a third of $\log M/M$ and decays cleanly, exactly as Proposition 5 predicts. The arithmetic tail is two to three orders of magnitude larger and does not decay monotonically at all. Whatever c_M does over this range, it is doing it for arithmetic reasons.

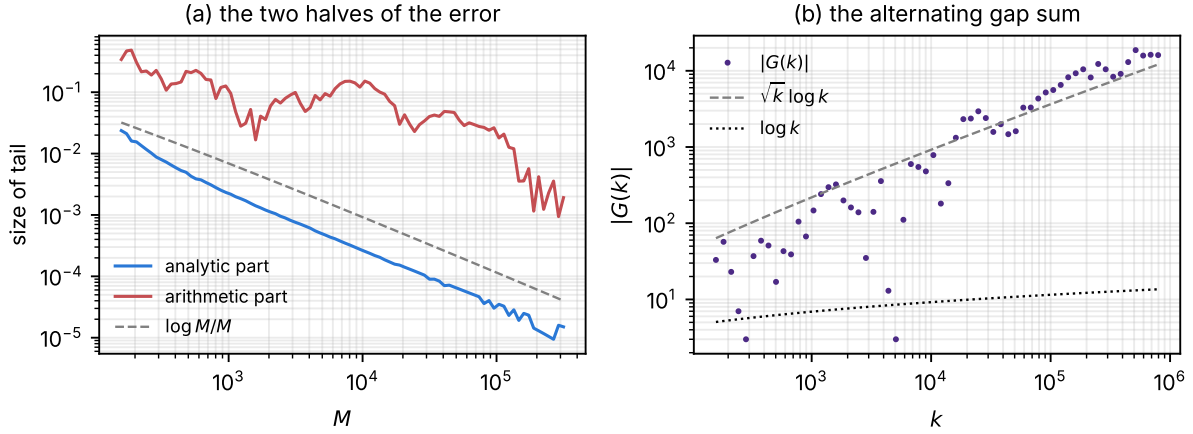


Figure 3: (a) The two tails of (21) at $x = 60.5$. The analytic part follows $\log M/M$; the arithmetic part is far larger and irregular. (b) The alternating gap sum $|G(k)|$, against $\sqrt{k} \log k$ and against $\log k$. Bounding $|G|$ by $\log k$ is what $c_M = O(1/M)$ would require.

9 Targeting a bound on G

Proposition 7 makes the rate of c_M a question about $G(k) = \sum_{j \leq k} (-1)^j g_j$ alone. This section reduces that question to a sharper one, records what the Riemann Hypothesis does and does not contribute, and gives the measured behaviour.

9.1 Reduction to gap fluctuations

Proposition 9. $|\sum_{j \leq k} (-1)^j \log p_j| \leq \log p_k$ for all $k \geq 1$.

Proof. Apply Lemma 4 with $a_j = \log p_j$. The right side of (19) is bounded by $\frac{1}{2} \sum_{j < k} |\log p_{j+1} - \log p_j|$, which telescopes to $\frac{1}{2}(\log p_k - \log p_1)$ because $\log p_j$ increases. Restoring the two boundary terms, each of modulus at most $\frac{1}{2} \log p_k$, gives the claim. $\square$

Writing $\varepsilon_j = g_j - \log p_j$ for the fluctuation of the gap about its mean size, this gives the unconditional reduction

$$G(k) = \sum_{j \leq k} (-1)^j \varepsilon_j + O(\log k). \quad (22)$$

Numerically the error term is not merely bounded by $\log p_k$ but sits at about half of it: at $k = 10^2, 10^3, \dots, 10^6$ it takes the values 2.60, 3.97, 5.26, 6.53, 7.77 against $\log p_k = 6.2, 8.9, 11.5, 14.1, 16.6$. So the smooth part of the gap contributes nothing, and the entire difficulty is the signed sum of the fluctuations.

9.2 Equivalence with the alternating sum of primes

Writing $\Sigma(k) = \sum_{i \leq k} (-1)^i p_i$ and reindexing,

$$G(k) = -2\Sigma(k) + (-1)^k p_{k+1} - 2, \quad (23)$$

so a bound on G is equivalent to one on the alternating sum of the primes themselves, and Lemma 4 identifies $G(k-1)$ with -2 times the half-endpoint-corrected $\Sigma(k)$. The same phenomenon therefore

appears twice in these notes: the interpolation error is a half-sum of gaps, and the gap sum is a half-sum of primes.

9.3 What the Riemann Hypothesis gives, and what it does not

The *unsigned* sum of fluctuations is completely standard. Since $\sum_{j \leq k} g_j = p_{k+1} - p_1$ and $\sum_{j \leq k} \log p_j = \vartheta(p_k)$,

$$\sum_{j \leq k} \varepsilon_j = p_{k+1} - \vartheta(p_k) - 2,$$

so on RH, using $\vartheta(x) = x + O(\sqrt{x} \log^2 x)$,

$$\sum_{j \leq k} \varepsilon_j \ll \sqrt{p_k} \log^2 p_k \ll \sqrt{k \log k} \log^2 k. \quad (24)$$

At $k = 10^6$ the left side is 6.03×10^3 against a bound of 1.06×10^6 , so this is far from tight, but it is unconditional in form and cheap.

It is also useless for our purpose. Splitting $A = \sum_{j \leq k, j \text{ even}} \varepsilon_j$ and $B = \sum_{j \leq k, j \text{ odd}} \varepsilon_j$, equation (24) controls $A + B$ while (22) asks for $A - B$. These are independent pieces of information, and no bound on the one constrains the other.

The obstruction can be stated more sharply. Since $(-1)^j = (-1)^{\pi(p_j)}$, the sum in (22) is a sum over primes weighted by the parity of the prime-counting function. Analytic methods reach π through ζ , which determines its size but carries no information about its parity, and there is no Dirichlet series with $(-1)^{\pi(p)}$ for coefficients that is amenable to the usual machinery. That is why the trivial bound $|G(k)| \leq p_{k+1} - 2 \ll k \log k$ has not been improved here.

9.4 The diffusive model, measured

If consecutive gap fluctuations are uncorrelated, G is a mean-zero random walk with step variance $\text{Var}(g)$, so $\text{Var}(G(k)) \sim k \text{Var}(g)$. This is testable without any modelling assumption, by measuring increments.

Over $k \in [2 \times 10^5, 10^6]$, where $\text{Var}(g) = 177.66$:

n	1	16	64	256	1024	16384
$\mathbb{E}[(G(k+n) - G(k))^2]/n$	431.1	186.5	184.9	186.8	178.3	169.7

The ratio is flat across four orders of magnitude in n , which is diffusive behaviour with no drift term; a linear drift would make the ratio grow like n . The $n = 1$ entry is not an exception: a single increment is $\pm g_{k+1}$, so its mean square is $\mathbb{E}[g^2] = \text{Var}(g) + \mathbb{E}[g]^2 = 177.66 + 253.44 = 431.10$, which is the tabulated value to five figures. For even n the mean cancels and the ratio drops to $\text{Var}(g)$.

Two further checks. The mean increment over that range is 0.00362 ± 0.02321 , consistent with zero, and the mean gap at even and odd index agrees to four decimals (15.9233 against 15.9160), so there is no parity bias at the level the data can resolve. And Figure 4(b) shows $|G(k)|$ staying inside the law-of-the-iterated-logarithm envelope $\sigma \sqrt{2k \log \log k}$, at a ratio rising from 0.13 at $k = 2^{12}$ to 0.83 at $k = 2^{19}$, which is a walk in a long one-signed excursion rather than a quantity with a trend.

One incidental observation: $\text{Var}(g)/\log^2 p_k = 177.66/253.44 = 0.70$ on this range, so the gaps are underdispersed relative to the exponential distribution that Cramér's model predicts. That

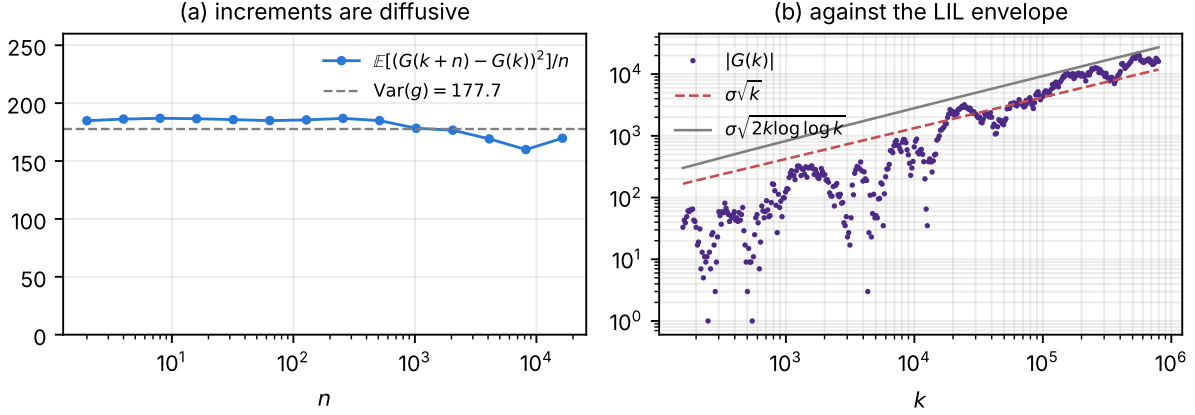


Figure 4: (a) Mean square increment of G per step, against step size, over $k \in [2 \times 10^5, 10^6]$. Flat means diffusive. The dashed line is the measured gap variance. (b) $|G(k)|$ against $\sigma\sqrt{k}$ and against the LIL envelope $\sigma\sqrt{2k \log \log k}$.

is consistent with the known failure of the naive Cramér model, and it makes the walk slightly slower than a first guess would suggest, but it does not change the exponent.

9.5 Conditional bound

Corollary 10. *If G obeys a law of the iterated logarithm with step variance $\sigma^2 = \text{Var}(g) \asymp \log^2 k$, then $G(k) \ll \sqrt{k} \log k \sqrt{\log \log k}$, and hence by Proposition 7,*

$$c_M \ll_x \frac{\log M \sqrt{\log \log M}}{\sqrt{M}}.$$

This is the target. It is consistent with every measurement above and it is what one should try to prove. What would suffice is square-root cancellation in $\sum_{j \leq k} (-1)^j \varepsilon_j$, which by (22) is the whole problem, and which by (23) is equivalent to square-root cancellation in the alternating sum of the primes. As far as these notes are aware that is open, and the parity-of- π obstruction above suggests it is not a small gap.

10 Richardson extrapolation to arbitrary order

By Proposition 7 the error splits as $r_M = r_\infty + (\text{analytic}) + (\text{arithmetic})$, and only the analytic part admits an expansion $a_1/M + a_2/M^2 + \dots$. Extrapolation in $1/M$ can therefore cancel that part and nothing else, which is worth keeping in view when reading the table below. Sampling at $M_i = N \cdot \rho^i$ for $i = 0, \dots, q$ and writing $t_i = 1/M_i$, evaluation of the degree- q interpolating polynomial in t at $t = 0$ gives Lagrange coefficients

$$\prod_{j \neq i} \frac{-t_j}{t_i - t_j} = \prod_{j \neq i} \frac{-\rho^{-j}}{\rho^{-i} - \rho^{-j}} = \prod_{j \neq i} \frac{1}{1 - \rho^{j-i}},$$

so that for any order

$$R_q(x) = \sum_{i=0}^q \lambda_i r_{N\rho^i}(x), \quad \lambda_i = \prod_{\substack{j=0 \\ j \neq i}}^q \frac{1}{1 - \rho^{j-i}}. \quad (25)$$

With $\rho = 2$ the first few coefficient sets are (1) , $(-1, 2)$, $(\frac{1}{3}, -2, \frac{8}{3})$ and $(-\frac{1}{21}, \frac{2}{3}, -\frac{8}{3}, \frac{64}{21})$, each summing to 1.

Measured against the limit, averaging $|\text{error}|$ over $x = 10.5, 25.5, 60.5, 125.5$:

primes used	N	$q = 0$	$q = 1$	$q = 2$	$q = 3$
1024	128	0.12144	0.06042	0.02149	0.00206
1000	250	0.08055	0.03586	0.00082	0.00194
2000	500	0.05820	0.00841	0.00179	0.04374
16000	1000	0.03330	0.00121	0.03805	0.09699

Order 2 with 1000 primes reaches 8×10^{-4} , a hundredfold gain. Beyond order 3 the extrapolation degrades, and the last row shows the counterintuitive consequence that with 16000 primes $q = 2$ is worse than $q = 1$.

Section 8 accounts for this. Richardson removes the analytic part, which Proposition 5 places at $O(\log M/M)$ and which the table at the end of that section shows sitting near 3×10^{-5} by $M = 10^5$. What remains is the arithmetic part, which is not a power series in $1/M$ at all and which the same table shows at 2×10^{-2} there. Once the analytic part has been cancelled, adding orders can only take differences of the arithmetic part, and differencing a random-walk-like quantity amplifies it by the size of the extrapolation coefficients, which for $\rho = 2$ grow like 2^q . Two facts follow, both visible in the table: the gain saturates near 10^{-3} rather than continuing, and more primes do not help once you are at the floor, because the floor is set by G and not by M alone.

11 Results

The limit values, computed as means over $M \in [3 \times 10^5, 1.03 \times 10^6]$ with a spread of 5.4×10^{-3} across that block:

x	$r_\infty(x)$	bracketing primes
2.5	4.130464	3, 5
3.5	5.675085	5, 7
10.5	30.021071	29, 31
20.5	71.703162	71, 73
60.5	281.736802	281, 283
125.5	696.003307	691, 701
200.5	1226.246641	1223, 1229

Richardson extrapolation in $1/M$ from the pair $M = 10^5, 10^6$ gives 281.736896 at $x = 60.5$ against the block mean 281.736802, agreeing to four decimals by an independent route. Figure 5 plots the limit function.

Two consistency checks are worth recording. The centred-window construction of Section 4, using only 250 primes, gave 695.840 at $x = 125.5$, which differs from the limit by 0.16; that is

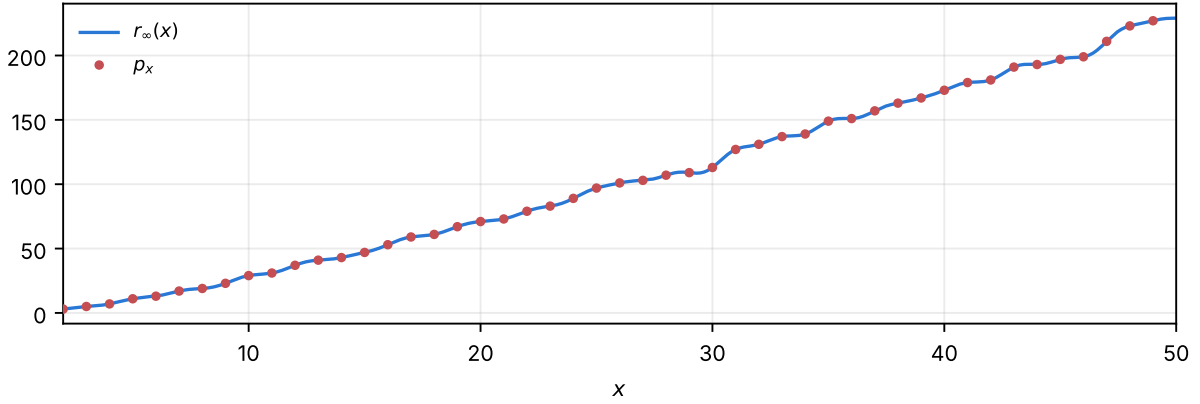


Figure 5: The limit function r_∞ on $2 \leq x \leq 50$, with the primes marked. It is C^∞ , interpolates every p_k exactly, and is determined by the prime sequence with no free parameters.

inside its own $O(h^{-1/2})$ error bar. And the limit is well behaved at the left edge, where every window-based method fails for want of room to centre: $r_\infty(2.5) = 4.130$ and $r_\infty(3.5) = 5.675$ both lie inside their bracketing pairs.

12 What is and is not established

Established. The Newton series (1) diverges at every non-integer x , with the mechanism and rate of Section 2. Recentring gives convergence at rate $O(h^{-1/2})$, by the cancellation (7). The barycentric $d = 1$ scheme has a limit as $M \rightarrow \infty$; the numerator converges only because of the endpoint halving, and Lemma 4 shows the halved sum is identically a half-sum of first differences. The approach to the limit is $c_M \sin(\pi x)$, verified across eight abscissas to five decimals. Proposition 5 bounds the analytic part of c_M by $O(\log M/M)$ unconditionally, using only $p_k \ll k \log k$; Proposition 6 bounds the arithmetic part by $O(\Phi(M)/M)$ for any regular majorant Φ of the alternating gap sum $|G|$.

Refuted. $c_M = O(1/M)$, asserted in an earlier version of these notes on the strength of a pointwise log-log fit. By Proposition 7 it would force $G(k) = O(\log k)$, and $|G(2^{18})| = 1.23 \times 10^4$ against $\log 2^{18} \approx 12.5$. The fit that produced it was computed with an estimator that is invalid for a quantity containing a random-walk term.

Not established. There is no uniqueness theorem behind any of this. The construction is parameter-free once $d = 1$ and $M \rightarrow \infty$ are fixed, but that makes it a canonical *recipe*, not a proven canonical function; a different equally parameter-free scheme could land elsewhere. Carlson's theorem cannot arbitrate, because r_∞ is a limit of rational functions with poles in $\mathbb{C}$, not an entire function of exponential type, so the type- π uniqueness machinery does not apply to it. The true size of $G(k)$ is unknown. Section 9 reduces it to square-root cancellation in $\sum_{j \leq k} (-1)^j (g_j - \log p_j)$, equivalently in the alternating sum of the primes, and shows that the Riemann Hypothesis does not reach it, since RH controls the unsigned sum $A + B$ while the question is about $A - B$. The measured behaviour is diffusive with no drift, which supports Corollary 10, but a measurement

over six decades is not a proof. Unconditionally we have only $c_M \ll \log M$, which is vacuous. And the extrapolation floor near 10^{-3} is an empirical observation from four abscissas, not a theorem.

A recurring number. At 250 primes, six unrelated routes gave the same uncertainty of roughly 0.1: the truncation residual of the centred window, the switch-point jump (8), the partition-of-unity blend envelope, the disagreement between the barycentric and window constructions, the sensitivity to the centring offset, and the dependence on M . That is not a defect of any one method. It is how much the interpolated value is determined by that much data.

A Implementation

Desmos. With the prime list $\mathbf{p}$ and $M = \text{length}(\mathbf{p})$, $K = [1, \dots, M]$:

$$\begin{aligned} u(n) &= \{K \leq n : 1, 0\} (-1)^K (2 - \{K = 1 : 1, 0\} - \{K = n : 1, 0\}), \\ g(x, n) &= \text{total}\left(\frac{u(n)p}{x-K}\right) / \text{total}\left(\frac{u(n)}{x-K}\right), \\ L(i) &= \prod_{j=0}^q \{j = i : 1, (1 - 2^{j-i})^{-1}\}, \quad R(x) = \sum_{i=0}^q L(i) g(x, N \cdot 2^i), \end{aligned}$$

with $N = \text{floor}(M/2^q)$. Note that $g(x, n)$ is meaningful only for x inside $[1, n]$; taking N too small relative to the plotted range produces errors in the hundreds rather than the thousandths.

Numerics. The difference tables were computed in exact integer arithmetic and evaluated with `mpmath` at 50–80 decimal digits, which is necessary: the centred sum at $h = 248$ cancels about 150 digits, and double precision fails there silently, returning a plausible but wrong curve. The barycentric computations are well conditioned and were done in double precision, with r_M for all M obtained in one pass by cumulative sums over a sieve of 1 031 130 primes below 1.6×10^7 .